Sri Eshwar College of Engineering
Coimbatore | Tamilnadu
An Autonomous Institution
Affiliated to Anna University, Chennai

Department of Mathematics

Assignment - Question Paper Pattern and Evaluation Rubrics

Course Code and Title : **U23MA204 - DISCRETE MATHEMATICS**

Year / Semester : **II / III**

Batch : **2025-2029**

Name of the Student : **Branch:**

Roll No :

Evaluation Rubrics :

Assignment Date:	**Last Date for Submission:**		
Rubric Criteria	**Levels of Assessment**		
	Inadequate (Below Standard)	**Adequate (Meets Standard)**	**Exemplary (Above expected Standard)**
Content **(30 Marks)**			
References **(10 Marks)**			
Presentation and Neatness **(10 Marks)**			
Total Score (Maximum-50 Marks)			

CO. No.	Course Outcome	BTL
U23MA204.1	Apply the fundamental concepts of logics and inference theory in proving and testing the logics.	K3
U23MA204.2	Operate counting problems by using the appropriate elementary counting techniques.	K3
U23MA204.3	Utilize induction techniques and generating functions to solve mathematical statements.	K4
U23MA204.4	Analyze the types of circuits in a graph, the existence of isomorphism and sketch the Euler and Hamiltonian paths and circuits in a graph.	K3
U23MA204.5	Validate the congruence using the classical theorems and can perform the divisibility test using congruence.	K5

S.No	Questions	CO	BT
1	Write a python program to prove conclusions by using inference rules: (i) $p \rightarrow q, q \rightarrow r, p \Longrightarrow r$ (ii) $p \vee q, \neg p \Longrightarrow q$.	CO1	K3
2	An IoT device sends an alert when Temperature > 45°C, Smoke detected, Gas leakage detected **Tasks** i. Write the logical proposition. ii. Simplify the proposition. iii. Verify logical equivalence using Boolean algebra. iv. Generate the truth table using Python.	CO1	K3
3	A drawer contains black, blue and red socks. Find the minimum socks to guarantee two of the same color. Write a python program.	CO2	K3
4	In a software company, Employees skilled in AI = 75, Employees skilled in Cloud = 68, Employees skilled in Cyber Security = 60, AI & Cloud = 28, AI & Cyber Security = 22, Cloud & Cyber Security = 18, and all three = 10. Write a Python program to calculate the number of employees possessing at least one skill.	CO2	K3
5	How many circular arrangements are possible for 8 people? Write a python program.	CO3	K3
6	Write a Python program to generate the first n terms of a sequence defined by the recurrence relation $y(n)=2y(n-1)$, $y(0)=3$ with a given initial value.	CO3	K3

7	Consider a social media platform where 12 users are connected through friendships. **Tasks** i. Represent the friendship network using a graph. ii. Construct its adjacency matrix. iii. Identify users having the maximum degree. iv. Determine whether the network is connected. v. Find all possible paths between User A and User H. vi. Suggest how influencers can spread information efficiently. vii. Write a python program.	CO4	K3
8	A tourist wants to visit every tourist place exactly once. Locations: A, B, C, D, E, F Roads: AB, AC, BC, CD, DE, EF, FA, BD, CE **Tasks** i. Draw the graph. ii. Find one Hamiltonian Circuit. iii. Explain why it satisfies the definition. iv. Write a python program	CO4	K3
9	Write a Python program to determine whether a resource allocation problem represented by the equation 35x + 20y = 240 has integer solutions. Display all valid non-negative solutions and explain how such equations are used in resource allocation	CO5	K3
10	Write a Python program that uses Wilson's theorem to determine whether the number 17 is prime.	CO5	K3

CO	PO 01	PO 02	PO 03	PO 04	PO 05	PO 06	PO 07	PO 08	PO 09	PO 10	PO 11	PSO 01	PSO 02
U23MA204.1	3	3	1	2	2	1	-	-	-	-	2		
U23MA204.2	3	3	1	1	2	1	-	-	-	-	2		
U23MA204.3	3	3	1	1	2	1	-	-	-	-	2		
U23MA204.4	3	3	2	2	2	1	-	-	-	-	2		
U23MA204.5	3	3	2	2	2	1	-	-	-	-	2		
Average	3	3	1	2	2	1	-	-	-	-	2		

"3"—High, "2"—Medium, "1"—Low, "-"—No Correlation

Faculty **HoD**